Checkpoint 2.10 — Automated Testing & Quality Assurance

NovaTech Industries | Intelligent Procurement & Logistics Management Platform | Python 3.12

Objective

Verify the Checkpoint 2.9 supplier evaluation engine using isolated temporary CSV fixtures. The tests invoke the actual engine without changing existing synthetic or analysis data.

Setup and execution

Save **test_supplier_evaluation_v2.py** in the **tests/** folder. From the project root, run:

```
python3 -m unittest discover -s tests -p 'test_*.py' -v
```

Reference result: 13 tests passed against the supplied Checkpoint 2.9 engine in an isolated environment.

Test coverage

Test	Expected behavior
Eligible quotes and ranking	Correct eligible status and top-ranked result
Weighted scoring	Cost 40%, delivery 25%, quality 20%, reliability 15%
Preliminary block	Upstream expiry or other block remains blocked
Invalid compliance	Nonvalid category approval blocks
Insufficient history	Review required, no invented score
Missing foreign key	Reject missing supplier performance
Duplicate ID	Reject duplicate quotation identifier
Invalid cost	Reject zero or negative cost
Invalid quality	Reject score above 100
Snapshot mismatch	Reject inconsistent reporting dates
Report before request	Verify current engine date guard
Exact scoring tie	Deterministic quotation-ID tiebreak
No automatic approval	Ranking never approves a purchase

Worked scoring example

S1: EUR 120, 5 days, quality 98, reliability 96. S2: EUR 100, 10 days, quality 92, reliability 84. S1 cost score = 83.33 and delivery score = 100; weighted score = **92.33**. S2 weighted score = **83.50**. Top-ranked status still requires human approval.

Known limitations

Point-in-time controls: A reporting-date check is not sufficient to prove source evidence existed at the original decision time. This is retrospective training analysis. Add event-level cutoffs and explicit evaluation timestamps before live procurement use.

Upstream validation: Expiry and inactive supplier logic lives in the earlier eligibility script. This suite checks that its BLOCKED results are honored, not that the upstream module itself is correct.

Policy: Evidence thresholds and tie-breaking rules remain configurable pending formal approval.

Practical exercise and acceptance criteria

Run all tests; deliberately alter a test fixture and observe the failure; restore it and rerun. Acceptance: all 13 tests pass, original CSVs remain unchanged, insufficient history requires review, blocked quotations are not ranked, and no purchase is automatically approved.

Next: Checkpoint 2.11 — Point-in-Time Evidence Controls & Expanded Integration Testing.